Sunlight-Robust Quadruped Parkour via Adaptive Depth-Corruption Training of World Models

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Fig. 1: Raw-depth quadrupedal parkour under head-on sun. The same policy traverses a gap (top) and a step (bottom) with a fixed camera configuration and no depth filtering. From left to right, each row shows the traversal, onboard RGB reference, raw depth input, and corresponding world model reconstruction; RGB is shown only for visual interpretation and is not used by the policy.

***Abstract*— Strong sunlight can remove obstacle boundaries and landing surfaces from active-stereo depth used for quadrupedal parkour. However, exposure to severe episode-persistent depth corruption from the start of training can impede policy learning. We present a structured, adaptive depth-corruption training framework for filter-free parkour directly from raw depth with a single policy and fixed camera configuration. The framework samples an episode-level corruption regime for each environment and adapts the corruption range after each rollout based on traversal progress and relative regime strength. We train a recurrent world model with clean-target denoising and clean–corrupted latent alignment under these regimes. Under the same training budget, simulation ablations show that performance-driven domain expansion achieves the highest traversal success under severe corruption and the smallest variation across seeds among the four evaluated training strategies. On a Unitree Go1, the learned policy traverses a 70 cm gap and 45 cm step across four outdoor illumination conditions. Under head-on sun, the policy traverses both obstacles, whereas none of the three evaluated baselines traverses either obstacle. The resulting policy extends the evaluated operational range of raw-depth parkour to head-on sun without deployment-time retuning. Videos and code are available at `https://dawnpp-parkour.github.io/`.**

I. Introduction

Quadrupedal parkour requires reliable perception of obstacle geometry and landing surfaces before takeoff. Learning-based systems achieve rapid rough-terrain locomotion [1] and egocentric-depth parkour with learned skills, state estimation, and navigation [2]–[9]. However, their parkour performance depends on deployment-time observations that preserve task-critical geometry.

Many systems use an Intel RealSense D435 or D435i active-stereo camera [2]–[6], [8]–[11]. Sunlight can overwhelm the projected infrared pattern and sharply reduce valid depth [12]–[15], while range, incidence angle, surface properties, and camera configuration further affect recovered geometry [16]–[18]. Conventional rendering does not accurately reproduce these active-stereo effects and thus creates an optical sensing domain gap [19]. Such structured degradation can erase an obstacle boundary and corrupt approach, takeoff, and landing decisions even when the robot and obstacle remain unchanged.

Deployment pipelines therefore crop, fill, or filter depth [3]–[5], [8], [20], yet favorable parameters vary with illumination, viewpoint, range, and surface appearance [14], [18], [21]. DAWN [22] removes external filtering through clean-depth reconstruction and recurrent latent alignment. Building on DAWN, we train the recurrent world model and policy under severe, episode-persistent depth corruption.

The central question is how training should progress from weaker to stronger corruption. Within the common training budget, max-strength sampling and full-range sampling underperform performance-driven expansion at the training maximum and at a held-out severity. Adaptive domain randomization and self-paced learning instead adjust training distributions according to learner competence [23]–[27]. For depth-based parkour, we base domain expansion on policy

performance under the sampled persistent regime rather than elapsed training or exposure alone. The update accounts for both traversal progress and relative regime strength because the same traversal outcome carries different information under weak and strong regimes.

We present a *structured, adaptive depth-corruption training framework* for vision-based parkour. The framework combines episode-persistent corruption regimes with performance-driven domain expansion. Regime parameters remain fixed within an episode, while traversal progress and relative regime strength determine the corruption ranges for subsequent episodes. We retain DAWN's clean-target reconstruction and latent-alignment objectives for the recurrent world model.

On a Unitree Go1, our system traverses a 70 cm gap and a 45 cm step from raw depth across four outdoor illumination conditions with a single policy, a fixed camera configuration, and no depth filtering. Under head-on sun, our system traverses both obstacles, whereas all three baselines and the variant without sunlight-related corruption fail. Our ablations isolate the effects of performance-driven domain expansion and sunlight-related corruption.

Our main contributions are:

- We develop a *structured, adaptive depth-corruption training framework* for vision-based parkour. The framework trains a recurrent world model and policy under episode-persistent corruption regimes constructed from depth-failure families.
- We introduce a *performance-driven domain expansion mechanism* that updates each environment's sampling range based on traversal progress and relative regime strength. This mechanism achieves higher traversal success at the training maximum and the held-out severity than max-strength sampling and full-range sampling within the same training budget.
- We extend the operational range of real-world raw-depth parkour to head-on sun. Our system traverses a 70 cm gap and a 45 cm step with a single policy, a fixed camera configuration, and no depth filtering.

II. Related Work

A. Visual Legged Locomotion

Parallel training and terrain curricula support sim-to-real locomotion [1]; proprioception-only controllers use implicit terrain representations [28] or observation–action histories [29]. Vision-conditioned controllers use depth policies [2], [3] and delay randomization [20]. Parkour systems distill specialized skills [4] or privileged inputs [5]; others use implicit–explicit estimation or skill libraries [6], [7]. WMP [8] learns recurrent visual dynamics without online planning; DAWN [22] processes raw D435i depth without manually tuned filters using clean-target denoising and clean–corrupted latent alignment. We retain DAWN's objectives but target sustained depth loss under head-on sun in a single active-stereo stream.

TABLE I: Learning designs and deployment outcomes. Depth denoising and domain expansion are learning designs; no sensing retuning and head-on sun parkour are outcomes from our evaluation.

	WMP [8]	Extreme Parkour [5]	DAWN [22]	**Ours**
Depth denoising	×	×	✓	✓
Domain expansion	×	×	×	✓
No sensing retuning	×	×	✓	✓
Head-on sun parkour	×	×	×	✓

B. Depth Degradation and Robust Learning

Active-stereo depth quality varies with range and incidence angle [16] and with sensor choice and scene content [17], [18]. Sunlight can overwhelm projected patterns [12], [14]; active-stereo failures can produce spatially structured missing depth [13]. Simulation and mixed-domain learning address active-stereo transfer [19], [30]. Depth restoration recovers geometry from synthetic corruptions [31]; denoising autoencoders learn representations robust to corrupted inputs [32]. Benchmarks evaluate robustness across corruption types and severity levels [33], [34], but perception metrics alone do not establish closed-loop traversal under persistent failures. Control-oriented methods suppress nuisance information [35] or integrate observation histories [8], [36]. Locomotion systems use redundant estimators [37], generalized depth models [10], corrupted-depth training [11], or stereo-artifact simulation [38]. We train the recurrent world model and policy under episode-persistent corruption regimes.

C. Domain Randomization and Curriculum Learning

Domain randomization targets appearance and dynamics [39]–[41]; its support and parameterization affect transfer [42]–[44]. SimOpt and BayesSim adapt simulators from real trajectories [45], [46]; active domain randomization selects informative environments [26]. Automatic domain randomization expands bounds after success [23]; self-paced and demonstration-based curricula adapt task distributions [24], [25], while TeachMyAgent benchmarks such methods [27]. Terrain curricula use performance [1], [5], [8]. Our domain expansion mechanism adapts the active corruption domain based on traversal progress, with relative regime strength gating expansion. Table I contrasts learning designs with deployment outcomes evaluated in this study.

III. Method

A. Task and System Overview

Sunlight-induced depth degradation varies with illumination, viewpoint, range, and surface properties. Rather than reproduce the full photometric image-formation process, we train the policy with synthetic corruptions that remove or distort task-relevant terrain information. Our framework composes depth-corruption operators into episode-persistent regimes and adapts their sampling ranges using traversal feedback.

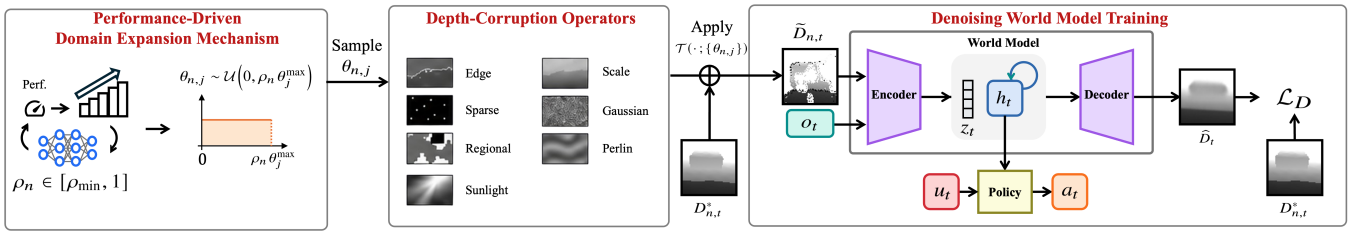


Fig. 2: Training pipeline. Traversal feedback updates the corruption-domain scale ρ_n, which sets the sampling ranges for $\theta_{n,j}$. The ordered operators map clean depth $D^*_{n,t}$ to corrupted input $\widetilde{D}_{n,t}$. The world model encodes $\widetilde{D}_{n,t}$ together with o_t; its decoder is trained to reconstruct $D^*_{n,t}$ using $\mathcal{L}_D$. Its recurrent state h_t conditions the policy.

We address partial observability with a recurrent world model that integrates corrupted depth observations over time. The deployment actor receives the latest deterministic recurrent state h_t, proprioceptive and previous-action history u_t, and motion command c_t:

$$a_t \sim \pi_\psi(\cdot \mid h_t, u_t, c_t), \tag{1}$$

where a_t is a joint-position target. A denoising world model maps the resulting corrupted depth stream to h_t, which conditions the policy trained with proximal policy optimization (PPO) [47]. Figure 2 summarizes the training pipeline.

B. Depth-Corruption Formulation

Each operator defines a *depth-failure family* through its magnitude and stochastic variation. Their sequential application models depth-value perturbations, sparse and edge-related failures, and regional missing depth. Let $\mathcal{T}$ denote the ordered sequence of operators and $\mathcal{T}_\ell$ its ℓth mapping. We omit the episode and time indices in this subsection and write

$$D^{(0)} = D^*, \qquad D^{(\ell)} = \mathcal{T}_\ell\left(D^{(\ell-1)}\right), \qquad \widetilde{D} = D^{(|\mathcal{T}|)}. \tag{2}$$

D^* is clean simulated depth, $D^{(\ell)}$ is the depth after applying the ℓth operator $\mathcal{T}_\ell$, and $\widetilde{D}$ is the final corrupted observation. Table II lists the operators in their application order, and Figure 3 shows their cumulative effect on one simulated scene. In the operator expressions below, D and D' denote the current input and output, respectively, and $\odot$ denotes elementwise multiplication.

The depth-scale operator applies a global scale perturbation:

$$R = (1 + r^{\text{scale}}\theta^{\text{scale}})D. \tag{3}$$

The sign $r^{\text{scale}} \in \{-1, +1\}$ determines the direction of the scale perturbation. The output R becomes the working depth for the next operator and remains the reference for range- and edge-dependent quantities. We normalize R over $[d_{\min}, d_{\max}] = [0.28, 2.0]$ m and clip the result to $[0, 1]$, denoting it by $\bar{d}$.

The Gaussian and Perlin operators add range-dependent unstructured and spatially correlated perturbations:

$$\begin{aligned} D'_{\text{g}} &= D + (\sigma_0 + \sigma_2 \bar{d}^2) \odot \varepsilon^{\text{g}}, \\ D'_{\text{p}} &= D + (A_0 + A_2 \bar{d}^2) \odot P. \end{aligned} \tag{4}$$

The field ε^{g} has independent standard Gaussian entries, and P is a normalized multi-octave Perlin field with $|P| \le 1$.

Edge dropout and sparse-pixel failures remove or saturate isolated measurements:

$$\begin{aligned} D'_{\text{edge}} &= (1 - M_{\text{edge}}) \odot D, \\ D'_{\text{sparse}} &= (1 - M_0 - M_{\text{sat}}) \odot D + d_{\max} M_{\text{sat}}. \end{aligned} \tag{5}$$

The binary mask M_{edge} selects edge-related dropout, while the disjoint masks M_0 and M_{sat} select zero and saturation failures on valid pixels. The probabilities p_{edge}, p_0, and p_{sat} control these masks, and the edge-related quantities are computed from R.

Regional operators remove coherent areas of terrain evidence:

$$\begin{aligned} D'_{\text{reg}} &= (1 - M_{\text{reg}}) \odot D + d_{\text{fill}} M_{\text{reg}}, \\ D'_{\text{sun}} &= (1 - M_{\text{sun}}) \odot (D + \sigma_{\text{sun}} B_{\text{sun}} \odot \varepsilon^{\text{sun}}) + d_{\text{fill}} M_{\text{sun}}. \end{aligned} \tag{6}$$

The structured-region mask M_{reg} is generated from a spatially correlated field, whereas M_{sun} selects a directional region. The area fractions α_{reg} and α_{sun} set the support sizes, and p_{reg} and p_{sun} set the respective boundary-failure probabilities. The binary mask B_{sun} selects surviving boundary pixels adjacent to the sunlight-related missing region, with $B_{\text{sun}} \odot M_{\text{sun}} = 0$. The field ε^{sun} has independent standard Gaussian entries, and both regional operators use $d_{\text{fill}} = d_{\max}$. We preserve the encoding of originally invalid pixels and clip the final output to the configured depth range. These operators represent terrain-information failures rather than the photometric formation of sunlight artifacts or physical illuminance.

C. Episode-Persistent Corruption Regimes

Let n index episodes within each parallel environment; sampling and domain updates are independent across environments, so we omit the environment index. Let $D^*_{n,t}$ be the clean simulated depth target. Applying the operator sequence produces the corrupted depth observation $\widetilde{D}_{n,t}$ received by the world model encoder. An *episode-persistent corruption regime* consists of the sampled magnitudes, signs, and persistence durations. The symbol $\theta_{n,j}$ denotes magnitude parameter j sampled for episode n. The index j enumerates the nonnegative magnitude parameters listed in Table II.

Let $[0, \theta_j^{\max}]$, with $\theta_j^{\max} > 0$, denote the full configured range of parameter j. If the full range is available, the

TABLE II: Depth-corruption operators. Operators are listed in application order with their full ranges and within-episode schedules.

Operator	Full range ($\rho_n = 1$)	Within-episode schedule
Depth-scale perturbation	$\theta^{\text{scale}} \le 0.10$	None
Distance-dependent Gaussian noise	$\sigma_0 \le 0.03\,\text{m}$; $\sigma_2 \le 0.07\,\text{m}$	ε_t^{g}: every update
Multi-octave Perlin bias	$A_0 \le 0.02\,\text{m}$; $A_2 \le 0.05\,\text{m}$	P_t: temporally evolving
Edge dropout	$p_{\text{edge}} \le 1$	Field: every 3–8 updates
Sparse-pixel failures	$p_0, p_{\text{sat}} \le 0.01$	Masks: every update
Structured regional failures	$\alpha_{\text{reg}} \le 0.25$; $p_{\text{reg}} \le 0.85$; $L_{\text{reg},n} \in [10, 20]$ updates	Support: every $L_{\text{reg},n}$; boundary: every update
Sunlight-related corruption	$\alpha_{\text{sun}} \le 0.50$; $p_{\text{sun}} \le 0.85$; $\sigma_{\text{sun}} \le 0.15\,\text{m}$; $L_{\text{sun},n} \in [20, 60]$ updates	Masks: every $L_{\text{sun},n}$; noise: every update

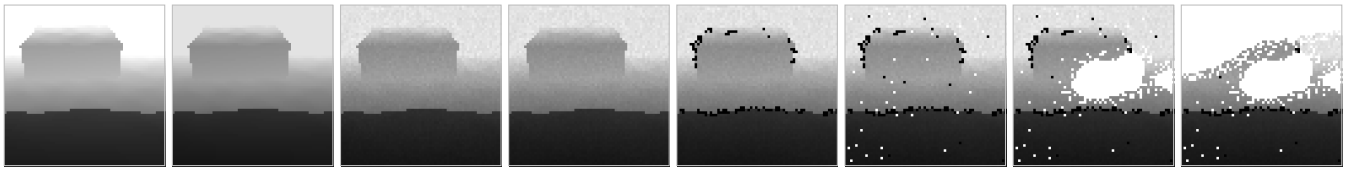


Fig. 3: Cumulative depth-corruption sequence. On a simulated gap scene, each panel adds the next of seven operators in the application order of Table II to the preceding output. All panels are 64×64 world model inputs.

reference sampling law is

$$\theta_{n,j} \sim \mathcal{U}\left(0, \theta_j^{\max}\right). \tag{7}$$

Unless stated otherwise, magnitude parameters are sampled independently. The depth-scale sign is sampled uniformly from $\{-1, +1\}$ at episode reset. Regional persistence durations are also sampled at reset and held for that episode.

The sampled magnitudes, sign, and persistence durations remain fixed during an episode. The operator-specific fields and masks vary according to the schedules in Table II and generate the stream $\widetilde{D}_{n,t}$. Episode persistence therefore specifies a sustained sensing condition; it does not repeat an identical depth image or mask.

D. Performance-Driven Domain Expansion

Full-range initialization can expose an untrained policy and world model to the largest corruption magnitudes. A narrow fixed range cannot cover the configured domain. The domain expansion mechanism instead adjusts the range based on performance rather than elapsed training.

The dimensionless corruption-domain scale ρ_n determines how much of each configured range is available for episode n:

$$\rho_n \in [\rho_{\min}, 1], \qquad \rho_0 = \rho_{\min} > 0. \tag{8}$$

We replace the fixed upper bound $\theta_j^{\max}$ in Eq. 7 with the active upper bound $\rho_n \theta_j^{\max}$:

$$\theta_{n,j} \sim \mathcal{U}\left(0, \rho_n \theta_j^{\max}\right). \tag{9}$$

Equation 9 defines the *active corruption domain*. Sampling one regime from it per episode constitutes depth-corruption domain randomization. At $\rho_n = 1$, the active range equals the full configured range, while episode-level magnitudes remain randomly sampled. For each regional operator, ρ_n expands the integer upper bound of the persistence range while its lower bound remains fixed. Thus, ρ_n controls the sampling support rather than realized image corruption or physical sunlight intensity.

Traversal progress alone can expand the range after success under an easy regime. We therefore measure how close the sampled magnitudes are to their current upper bounds. The index $j = 1, \ldots, J$ ranges over the sampled magnitudes of all operators except the depth-scale perturbation. The *relative regime strength* is

$$s_n = \frac{1}{J} \sum_{j=1}^{J} \frac{\theta_{n,j}}{\rho_n \theta_j^{\max}}. \tag{10}$$

The score $s_n \in [0, 1]$ is the mean normalized magnitude of the sampled non-scale components; it does not measure image-level corruption.

After each rollout, we compute traversal progress x_n and strength s_n to update ρ_{n+1}. The displacement x_n is measured from the start of the current terrain tile at episode termination, and L_{tile} is the tile length. The commanded forward velocity used in the terminal check is $v_{x,n}^{\text{cmd}}$, and T_{ep} is the maximum episode duration. The thresholds η_x, η_s, and η_p specify sufficient traversal, relative regime strength, and poor progress, respectively, and $\Delta\rho$ is the update step. We increase ρ_n when $x_n > \eta_x L_{\text{tile}}$ and $s_n \ge \eta_s$. We decrease ρ_n when $x_n < \eta_p |v_{x,n}^{\text{cmd}}| T_{\text{ep}}$ and $x_n \le \eta_x L_{\text{tile}}$. Otherwise, ρ_n remains unchanged. The bounded update is

$$\rho_{n+1} = \begin{cases} \min(1, \rho_n + \Delta\rho), & \text{increase}, \\ \max(\rho_{\min}, \rho_n - \Delta\rho), & \text{decrease}, \\ \rho_n, & \text{otherwise}. \end{cases} \tag{11}$$

Each environment applies this partial-traversal update independently at episode boundaries; evaluation instead uses

complete-course success. The resulting ρ_{n+1} sets the next episode's corruption ranges.

E. Denoising World Model Training

The domain scale changes the distribution of corrupted inputs, while the world model objective remains fixed. The corrupted input $\tilde{x}_t = (\widetilde{D}_t, o_t)$ and its paired clean input $x_t^* = (D_t^*, o_t)$ share proprioception and the motion command but differ only in depth. We omit the episode index; t indexes world model updates in this subsection and policy steps in control equations.

Following WMP [8] and DreamerV3 [48], we use a recurrent state-space model (RSSM) with parameters ϕ. The concatenated actions between successive world model updates are denoted by $\bar{a}_{t-1}$. We denote the combined deterministic and stochastic state by $b_t = (h_t, z_t)$. The corrupted path comprises four components:

$$\begin{aligned} \text{Recurrent model:} \quad & h_t = f_\phi(h_{t-1}, z_{t-1}, \bar{a}_{t-1}), \\ \text{Encoder:} \quad & z_t \sim q_\phi(\cdot \mid h_t, \tilde{x}_t), \\ \text{Dynamics predictor:} \quad & \hat{z}_t \sim p_\phi(\cdot \mid h_t), \\ \text{Depth decoder:} \quad & \widehat{D}_t = g_\phi(b_t). \end{aligned} \tag{12}$$

h_t is the deterministic recurrent state, z_t is the posterior stochastic state conditioned on the corrupted observation, and $\hat{z}_t$ is the prior prediction without the current observation. The current corrupted observation updates z_t and enters h_{t+1}; between world model updates, the actor reuses the latest h_t.

Following DAWN [22], the corrupted depth is the encoder input and its paired clean depth is the reconstruction target. The clean-target denoising loss over T world model updates is

$$\mathcal{L}_D = \frac{1}{T} \sum_{t=1}^{T} \left\| \widehat{D}_t - D_t^* \right\|_F^2 . \tag{13}$$

The shared RSSM maps x_t^* and $\tilde{x}_t$ to b_t^{clean} and b_t^{cor}, respectively. As in DAWN [22], a shared projection head maps these states. Projected outputs at the same batch–time index form positive pairs, and the rest form negatives in $\mathcal{L}_{\text{align}}$. The paths also share actions, reset flags, and camera alignment.

On the corrupted path, a vector decoder predicts o_t from b_t. We add its negative log-likelihood to Eq. 13 to form $\mathcal{L}_{\text{pred}}$. We combine $\mathcal{L}_{\text{pred}}$ with the standard RSSM dynamics and representation losses $\mathcal{L}_{\text{dyn}}$ and $\mathcal{L}_{\text{rep}}$ [48]:

$$\mathcal{L}_{\text{WM}} = \mathcal{L}_{\text{pred}} + \beta_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \beta_{\text{rep}}\mathcal{L}_{\text{rep}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}}. \tag{14}$$

The coefficients β_{dyn}, β_{rep}, and λ_{align} are nonnegative loss weights. Actor gradients do not pass through the detached deterministic state to the world model. The corrupted path supplies $\mathcal{L}_{\text{pred}}$, $\mathcal{L}_{\text{dyn}}$, and $\mathcal{L}_{\text{rep}}$, while the clean path is used only for alignment.

F. Policy Learning and Deployment

We train the actor in Eq. 1 from simulator rollouts using PPO [47]. The ratio w_t is the current-policy action probability divided by the behavior-policy action probability. $\hat{A}_t$ is the advantage estimate computed with generalized advantage estimation [49], and ϵ is the clipping bound. We maximize the clipped surrogate objective

$$J_{\text{PPO}}(\psi) = \mathbb{E}_t \left[\min\Big(w_t \hat{A}_t,\ \text{clip}(w_t, 1-\epsilon, 1+\epsilon)\, \hat{A}_t \Big) \right]. \tag{15}$$

An auxiliary head uses simulator ground-truth base velocity to supervise the history encoder during training.

During training, an asymmetric critic [50] additionally receives the privileged observation o_t^{priv}. Adversarial Motion Priors (AMP) [51] provide a style reward. We apply dynamics domain randomization [40] to mass, center-of-mass, friction, contact, and actuation parameters.

At deployment, measured real-world depth D_t^{meas} replaces simulated corrupted depth in q_ϕ, and the actor uses the latest h_t through Eq. 1. The corruption generator, clean path, projection and auxiliary heads, privileged inputs and targets, AMP discriminator, and domain expansion mechanism are training-only.

IV. Experiments

We evaluate outdoor traversal, training choices, reconstruction quality, and robustness limits.

A. Experimental Setup

Platform and training. We deploy on a Unitree Go1 with an Intel RealSense Depth Camera D435i and an onboard Jetson NX, and train in 4096 Isaac Lab [52] environments. The D435i streams 424×240 depth at 60 Hz. We crop and resize the depth to 64×64. The policy and RSSM operate at 50 and 10 Hz, respectively. Each configuration uses 5 seeds, 20,000 training iterations, and 200 evaluation episodes per seed and condition. The AMP discriminator supplies the style reward. We set $\beta_{\text{dyn}} = 0.5$, $\beta_{\text{rep}} = 0.1$, and $\lambda_{\text{align}} = 0.1$.

Dynamics and observation randomization. Dynamics randomization covers trunk mass and center of mass, contacts, joint friction, actuator gains, observation delays, and planar pushes. Table II gives the image-space ranges; these and all non-ablated settings are shared across configurations. The domain expansion settings are $\eta_x = 0.625$, $\eta_s = 0.50$, $\eta_p = 0.50$, $\Delta\rho = 0.1$, and $\rho_0 = \rho_{\min} = 0.1$. The domain update uses $L_{\text{tile}} = 16$ m and $T_{\text{ep}} = 20$ s. At episode reset, $L_{\text{reg},n}$ and $L_{\text{sun},n}$ are sampled uniformly from the inclusive integer ranges $\{10, \ldots, 10 + \lfloor 10\rho_n \rfloor\}$ and $\{20, \ldots, 20 + \lfloor 40\rho_n \rfloor\}$. Each support mask is redrawn after its sampled persistence duration.

Outdoor illumination. We evaluate shade, open daylight, side-on sun, and head-on sun. Shade excludes direct sunlight. Open daylight is unshaded daylight with measured illuminance below the ranges for side-on and head-on sun. Side-on and head-on denote the sun direction relative to traversal. Table III reports illuminance ranges in kilolux (klx), and Fig. 1 shows the scenes, traversals, and raw depth.

Courses and deployment configurations. The physical courses use the most difficult available movable obstacles: a Step (45 cm) and a Gap (70 cm). Obstacle geometry, surface material, commands, start and finish regions, and the policy checkpoint are held fixed within each comparison. Our

TABLE III: Real-world parkour success. Entries are successful trials out of 20 (%) on the hardest tested gap and step in each illumination condition. Ours and DAWN use fixed camera settings; WMP and Extreme Parkour use the best tested camera setting per condition.

Method	Shade < 10 klx		Open daylight 10–30 klx		Side-on sun 30–200 klx		Head-on sun 30–200 klx	
	Step	Gap	Step	Gap	Step	Gap	Step	Gap
Extreme Parkour	11/20 (55)	11/20 (55)	12/20 (60)	11/20 (55)	9/20 (45)	10/20 (50)	0/20 (0)	0/20 (0)
WMP	18/20 (90)	18/20 (90)	17/20 (85)	18/20 (90)	16/20 (80)	15/20 (75)	0/20 (0)	0/20 (0)
DAWN	18/20 (90)	18/20 (90)	19/20 (95)	17/20 (85)	18/20 (90)	18/20 (90)	0/20 (0)	0/20 (0)
Ours w/o sun corr.	19/20 (95)	18/20 (90)	18/20 (90)	19/20 (95)	18/20 (90)	18/20 (90)	0/20 (0)	0/20 (0)
Ours	**20/20 (100)**	**19/20 (95)**	**20/20 (100)**	**20/20 (100)**	**19/20 (95)**	**19/20 (95)**	**19/20 (95)**	**19/20 (95)**

method uses a single policy across both obstacles and all four conditions, without changing the camera settings or applying any depth filter.

Comparisons. We compare against Extreme Parkour [5], WMP [8], and DAWN [22], plus a variant that removes only sunlight-related corruption (Ours w/o sun corr.). DAWN and our variants retain one sensing configuration across conditions; WMP and Extreme Parkour use processed depth and their best tested condition-specific configuration.

Metrics. A real-world trial succeeds when the robot traverses the obstacle without falling. We report successes and attempts for every method–course–illumination combination. For each simulation seed and condition, we compute the success fraction over 200 episodes and report the mean and standard deviation of these fractions. For simulation evaluation, we define normalized corruption severity such that 0 denotes clean observations and 1 the final training-domain maximum; 1.25 is the held-out severity.

B. Real-World Parkour across Outdoor Illumination Conditions

Table III shows that Ours is the only evaluated method that traverses both obstacles under every illumination condition with a single sensing configuration. The distinction is clearest under head-on sun, where Ours achieves 95% success while the baselines and the variant without sunlight-related corruption fail. The ablated policy remains effective in the other conditions, showing that sunlight-related corruption training is specifically responsible for transfer to head-on sunlight. During training, this operator introduces directional missing regions and corrupted boundary measurements. The world model–policy system thereby learns to use the terrain evidence that remains instead of depending on complete depth measurements.

C. Effect of Adaptive Corruption Training

To assess performance-driven domain expansion and the contribution of sunlight-related corruption, we compare five configurations that share the same architecture, training budget, and evaluation protocol (Table IV). Maximum-strength regime sampling holds each nonnegative corruption magnitude at its upper bound, whereas Full-range regime sampling draws each episode's regime from the full configured domain from initialization. Fixed schedule expands the domain according to elapsed training, whereas Ours uses traversal feedback. Ours w/o sun corr. retains performance-driven expansion but omits sunlight-related corruption. All configurations retain the realization schedules of their included operators from Table II; only the regime magnitudes, domain-update rule, or included operators vary. All corrupted evaluations use the episode-persistent process defined in Sec. III-C.

TABLE IV: Simulation ablation results. Traversal success (%) is reported for clean observations, the training maximum, and the held-out severity. Each seed is evaluated over 200 episodes; values are the mean ± standard deviation across 5 seeds.

Variant	Clean	Training max.	Held-out
Max-strength sampling	100.0 ± 0.0	56.8 ± 7.1	46.3 ± 11.5
Full-range sampling	100.0 ± 0.0	85.6 ± 4.1	80.3 ± 6.2
Fixed schedule	100.0 ± 0.0	85.3 ± 7.5	77.2 ± 8.3
Ours w/o sun corr.	100.0 ± 0.0	55.9 ± 4.8	43.1 ± 7.5
Ours	**100.0 ± 0.0**	**97.3 ± 0.5**	**94.2 ± 1.3**

Table IV shows that all five configurations achieve 100% success under clean observations. At the training maximum, Ours achieves 97.3% success, compared with 85.6% for full-range sampling and 85.3% for the fixed schedule. At the held-out severity of 1.25, Ours retains 94.2% success with a standard deviation of 1.3 percentage points, whereas full-range sampling and the fixed schedule achieve 80.3% and 77.2%, respectively. Ours w/o sun corr. reaches 55.9% and 43.1% at the two corrupted evaluation levels. These results show that performance feedback improves traversal under strong corruption, while the no-sun ablation shows that domain expansion alone is insufficient.

D. World Model Reconstruction under Depth Degradation

Figure 4 shows that the world model retains the coarse obstacle layout, height transition, and landing region despite large missing regions in the corrupted input. Thin boundaries remain smoother than in the clean reference, but the geometry needed to distinguish the obstacle from its landing surface is preserved.

The clean reconstruction target makes terrain structure, rather than corruption-specific appearance, useful for minimizing the denoising loss. Latent alignment further brings

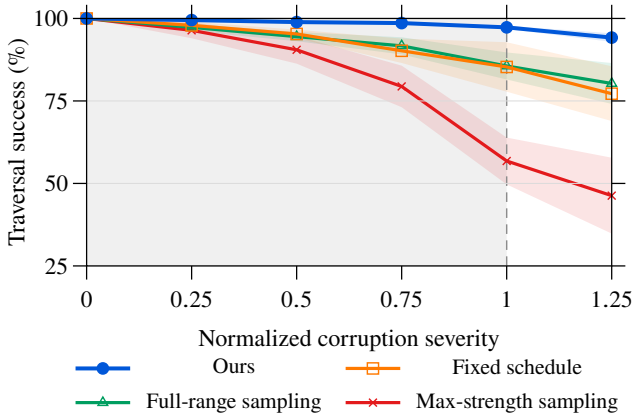


Fig. 5: Corruption severity sweep. Gray shading marks the training range. Each seed is evaluated over 200 episodes; curves and bands show the mean and ±1 standard deviation across 5 seeds.

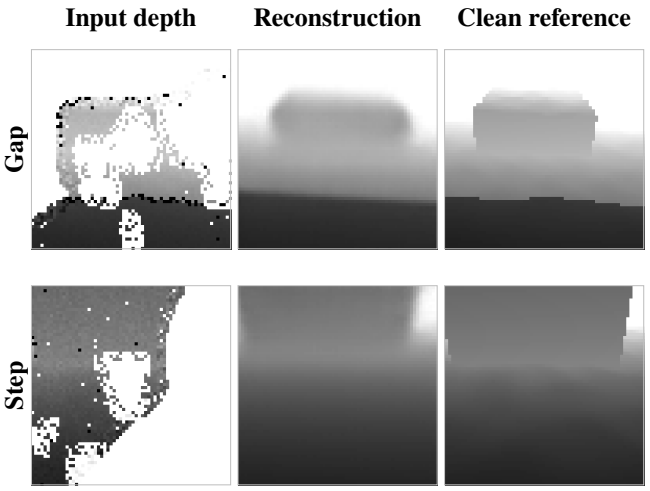


Fig. 4: Terrain reconstruction. Rows show simulated gap and step scenes; columns show corrupted input, world model reconstruction, and aligned clean reference. All panels use aligned crops and a common depth range.

the clean and corrupted states of the same scene together. The recurrent dynamics carry this terrain information across observations, and the actor receives the resulting recurrent state instead of the corrupted depth image itself. This representation allows the same policy to act even when the current depth frame contains incomplete obstacle geometry.

E. Performance versus Corruption Severity

Figure 5 shows that the methods perform similarly near the clean setting but separate as normalized corruption severity increases. Ours degrades gradually across the training range and retains high traversal success at the held-out severity. Fixed schedule deteriorates more rapidly beyond the training maximum. The benefit of performance-driven expansion therefore extends beyond the corruption range encountered during training and remains effective as the same depth-failure modes become stronger.

Performance-driven expansion improves traversal under strong corruption by adapting exposure to severe corruption based on policy performance, while the recurrent world model preserves terrain structure from incomplete depth. Together, these capabilities enable the full system to maintain high traversal success across and beyond the training range.

V. Conclusion

We presented a structured, adaptive depth-corruption training framework that composes depth-corruption operators into episode-persistent regimes. Performance-driven expansion improves traversal success under strong corruption. Denoising and latent alignment preserve terrain geometry in the recurrent world model. With a single policy, a fixed camera configuration, and no depth filtering, our system traverses a 70 cm gap and a 45 cm step across four outdoor illumination conditions. Our method is the only evaluated approach to traverse both obstacles under head-on sun.

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